PAPER_229: Pillars of Creation (Eagle Nebula M16) — MUGE with Decaying Photoevaporation Erosion Factor E(t)

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Session: 58 (grokshare8d951e12.txt extraction — Doc 7)

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Author: Daniel T. Murphy **Framework:** UQFF v4.8 (Star-Magic) **Session:** 58 (grok*share*8d951e12.txt extraction — Doc 7) **Date:** March 2026 **Classification:** Novel MUGE Term — Decaying Erosion Suppression (1 – E(t)) on Base Gravity **Status:** Proof-Quality Whitepaper

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Abstract

The Pillars of Creation in the Eagle Nebula (M16, NGC 6611, ~6,500 ly) are modelled with a 9-term MUGE incorporating a uniquely novel term: a decaying photoevaporation erosion factor $E(t) = E_0 e^{-t/\tau_e}$ applied to the base gravity as a multiplicative suppression $(1 - E(t))$. This is physically distinct from the Bubble Nebula (PAPER_221) which uses $(1 + E(t))$ for shell compression enhancement. The sign physically distinguishes systems where EUV radiation removes mass (erosion → less gravity) versus compresses it (shock → more gravity).

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1. Physical System

The Pillars are evaporating proto-stellar columns irradiated by the NGC 6611 O-star cluster:

Parameter	Value
Distance	~6,500 ly
Pillar height	~4–5 ly
M_{init}	$100 M_\odot$ (pillar stellar cores)
M_{gas}	$10{,}000 M_\odot$ (column gas mass)
r	5 ly
B	$1\ \mu$T
EUV source	NGC 6611 O-stars ($T_{eff} > 40{,}000$ K)
E_0	0.1 (10% peak erosion suppression)
τ_e	1 Myr

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2. Decaying Erosion Factor (Novel)

2.1 Definition

```
E(t) = E_0 e^{-t/\tau_e}
```

2.2 Application to Base Gravity

```
a_{base} = \frac{GM(t)}{r^2}(1 + H_0 t)\left(1 - \frac{B}{B_{crit}}\right)(1 - E(t))
```

At `t = 0`: `E = E_0 = 0.1` → base gravity suppressed by 10% (maximum erosion). At `t \gg \tau_e`: `E \to 0` → gravity recovers fully as the pillar material disperses or the EUV source weakens.

2.3 Physical Interpretation

The factor `(1 - E(t))` represents the fractional mass remaining in the gravitational region after photoevaporative mass loss. As EUV photons ablate the pillar surface, the effective gravitating mass decreases, reducing `a_{base}`.

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3. Sign Comparison: Erosion vs Compression

System	Erosion/Expansion Term	Sign	Physical Process
Pillars of Creation (PAPER_229)	`(1 - E(t))`, `E_0 = 0.1`	−	EUV ablation removes mass → less gravity
Bubble Nebula (PAPER_221)	`(1 + E(t))`, growing	+	Shock compression → more gravity
Orion Nebula (PAPER_xxx)	None	N/A	Young, pre-dispersal

The sign of the erosion factor is physically determined by whether the process removes or adds mass to the gravitating region.

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4. Canonical Result

At `t = 0.1` Myr with `M(0) = 100.1 M_\odot`:

```
a_{base} \approx \frac{G \times 1.989 \times 10^{32}}{(4.73 \times 10^{16})^2} \times (1 - 0.1 \times e^{-0.1}) \approx 5.93 \times 10^{-24} \times 0.905 \approx 5.36 \times 10^{-24} \text{ m/s}^2
```

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5. Simulation Parameters

Parameter	Value
`t_{canonical}`	`0.1` Myr
dt	`0.001` Myr
`M_{dot\_factor}`	`M_{gas}/M_{init} = 100`
`\tau_{SF}`	`5` Myr
`E_0`	`0.1`
`\tau_e`	`1` Myr

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6. Calculator Class

```
class PillarsOfCreationErosionMUGECalculator(_CP3Calculator):
"""PAPER_229: Pillars of Creation — 9-term MUGE, decaying erosion (1-E(t)) on base gravity"""
# Session 58 — grok_share_8d951e12.txt Doc 7
```

Located in `CondensedPhysics3.py` (Session 58).

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7. Conclusion

The decaying erosion factor `(1 - E_0 e^{-t/\tau_e})` is a genuinely novel MUGE term. Combined with the sign-comparison against the Bubble Nebula's `(1 + E(t))`, this establishes a physically rigorous erosion/compression taxonomy within the UQFF nebular MUGE family: negative sign for ablative processes, positive for compressive ones.

Source: grok*share*8d951e12.txt — Doc 7 (Pillars of Creation Erosion MUGE)

UQFF computed: MUGE buoyancy ratio U*bi/F*U = [SSq]■?■r■/GM = 5.7e-1■5.0e-4 = 2.85e-4; compressed MUGE baseline g = 5.4e-7 m/s■ at r_ISCO.
